

Smoking Model Methodology

We directly model smoking history as a discrete-time, age-indexed Markov Process with three states: Never (1), Current (2) and Former (3, absorbing). For each individual i , annual state transitions are simulated each year from age 12 to 75. The observed final state after each year contributes to the likelihood via forward recursion.

State and transitions

Allowed transitions per year t (age a) are:

- Initiation: $1 \rightarrow 2$
- Cessation: $2 \rightarrow 3$
- Former is absorbing: $3 \rightarrow 3$

Assuming a age-dependent annual Markov transition matrix:

$$P(a) = \begin{bmatrix} 1 - p_{12}(a) & p_{12}(a) & 0 \\ 0 & 1 - p_{23}(a) & p_{23}(a) \\ 0 & 0 & 1 \end{bmatrix}$$

Within a year, the state probability vector (s_1, s_2, s_3) is updated by:

$$\begin{aligned} s'_1 &= s_1(1 - p_{01}(a)) \\ s'_2 &= s_2(1 - p_{12}(a)) + s_1p_{01}(a) \\ s'_3 &= s_3 + s_2p_{12}(a) \end{aligned}$$

followed by a small numerical clamp to $[0, 1]$ and renormalization to preserve the simplex.

Grouping structure

Individuals are stratified by gender $\times$ race into $G = 2 \times K_{\text{race}}$ groups g . We include non-centered, group-level random intercepts for each transition:

$$z_g^{(01)}, z_g^{(12)} \sim \mathcal{N}(0, 1), \quad \tau_{01} > 0, \tau_{12} > 0.$$

Birth-cohort effects are incorporated using discrete 5-year year-of-birth categories $\text{yob_group}_i \in \{-3, -2, -1, 0, 1, 2, 3, 4, \dots\}$.

We center the cohort index at 1970 (i.e., $\text{yob_group}_i = 0$ for 1970–1974 births), reflecting the onset of major national tobacco-control efforts during the 1970s. Cohort effects act multiplicatively on the hazard through log-linear coefficients.

Annual transition probabilities (cloglog link)

We use the complementary log-log (cloglog) link to map linear predictors to annual transition probabilities, consistent with discrete-time hazards:

$$p = 1 - \exp(-\exp(\eta))$$

For numerical stability we compute p with a safeguarded routine using `expm1` and overflow guards.

- Initiation 1 $\rightarrow$ 2

$$\eta_{ig}^{(01)}(a) = \theta_0 + \tau_{01} z_g^{(01)} + \beta_{yob}^{(01)} \cdot yob_group_i + \left[-\frac{(a - 20)^2}{2s_{01}^2} \right],$$

$$p_{01}(a) = 1 - \exp(-\exp(\eta_{ig}^{(01)}(a))).$$

The bell term centers initiation propensity near age 20 with width s_{01} .

- Cessation 2 $\rightarrow$ 3

$$\eta_{ig}^{(12)}(a) = \gamma_0 + \tau_{12} z_g^{(12)} + \gamma_1 (a - 40) + \beta_{yob}^{(12)} \cdot yob_group_i,$$

$$p_{ig}^{12}(a) = 1 - \exp(-\exp(\eta_{ig}^{(12)}(a)))$$

The linear age slope $\gamma_1 > 0$ captures increasing cessation with age (centred at 40).

Likelihood

Assuming independence across individuals, the log-likelihood is:

$$l(\theta) = \sum_{i=1}^{\theta} \log(p_i(\theta) + 10^{-12})$$

Priors and constraints

We place weakly informative priors on interpretable scales (log-normal via normal priors on log) and standard normal priors on random-effect standards:

- Initiation:

$$\theta_0 \sim \mathcal{N}(\log 0.03, 0.1), \log \tau_{01} \sim \mathcal{N}(\log 3, 0.1), \log s_{01} \sim \mathcal{N}(\log 6, 0.1), \log \beta_{yob}^{(01)} \sim \mathcal{N}(\log 0.1, 0.5).$$

- Cessation:

$$\gamma_0 \sim \mathcal{N}(\log 0.1, 0.1), \log \gamma_1 \sim \mathcal{N}(\log 0.03, 0.5), \log \tau_{12} \sim \mathcal{N}(\log 3, 0.1), \log \beta_{yob}^{(12)} \sim \mathcal{N}(\log 0.1, 0.5).$$

- Random effects:

$$z_g^{(01)}, z_g^{(12)} \sim \mathcal{N}(0, 1).$$

- Transformed parameters enforce positivity:

$$s_{01}, \tau_{01}, \tau_{12}, \gamma_1 > 0.$$

Transition Rates Estimation in micro-level data

To estimate the transition rate parameters, we applied ADVI to approximate the posterior distribution by minimizing the K-L divergence between the true posterior and a variational family under an automatic differentiation framework. Learning rate $\epsilon = 0.05$, iter = 50,000 and 20,000 draws were generated from the variational posterior for diagnostics.

And then we used NUTS with warm-up being 2,000 and sampled 4,000 samples from the model.

DEMOS-Smoking History generator

We incorporated the estimated transition rates into DEMOS, our stochastic microsimulation platform. Because the overall initiation level and cohort gradients in the longitudinal study differ from those observed in the calibration targets used in DEMOS, we introduced scaling factors to reconcile these differences. Specifically, multiplicative adjustments were applied to (i) the baseline initiation annual probability $\lambda_{\text{init},g}$ and (ii) the cohort-specific slope for both gender, denoted as $\lambda_{\text{cohort},g}$, where $g \in \{0, 1\}$, 0 = male and 1 = female.

Notice

In practice, we noticed that DEMOS generated prevalence in Malay group still underestimated. Therefore, we also added two Malay-specific scale and cohort related slope, denoted as $\lambda_{\text{init},\text{malay}}$, $\lambda_{\text{cohort},\text{malay}}$.

To avoid overlap among targets, we choose our calibration targets as overall prevalence for male and female group from 2019 to 2024 together with corresponding 95% CIs.

Denote target prevalence for gender g and year y as $\hat{p}_{g,y}$ with 95% CI $[L_{g,y}, U_{g,y}]$. Then we convert each target to logit scale:

$$z_{g,y} = \text{logit}(\hat{p}_{g,y}), \quad \text{SE}_{g,y} \approx \frac{\text{logit}(U_{g,y}) - \text{logit}(L_{g,y})}{2 \times 1.96}.$$

Then we run DEMOS with $\underline{\lambda} = (\lambda_{\text{init},g}, \lambda_{\text{cohort},g}, \lambda_{\text{init},\text{malay}}, \lambda_{\text{cohort},\text{malay}})$ to get predictions $p_{g,y}(\underline{\lambda})$, then $z_{s,y}(\lambda) = \text{logit}(p_{g,y}(\lambda))$.

So ultimately we minimize weighted SSE using Nelder-Mead with respect to $\underline{\lambda}$:

$$\mathcal{L}(\underline{\lambda}) = \sum_{g \in \{0,1\}} \sum_y \frac{[z_{g,y}(\underline{\lambda}) - z_{g,y}]^2}{\text{SE}_{g,y}^2}.$$